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IoT Based Energy Meter Monitoring System

By
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FINAL YEAR DESIGN PROJECT REPORT

This Report Presented in Partial Fulfillment of the Requirements forthe **Degree of Bachelor of Science in Computer Science and Engineering**

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December 09, 2025

APPROVAL

This Project titled "**IoT Based Energy Meter Monitoring System**," submitted by **Md. Sakib Al-Hasan** to the Department of Computer Science and Engineering, Daffodil International University, has been accepted as satisfactory for the partial fulfillment of the requirements for the degree of B.Sc. in Computer Science and Engineering and approved as to its style and contents. The presentation has been held on **09-12-2025**.

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DECLARATION

We hereby declare that this project has been done by us under the supervision of **Dr. Fizar Ahmed, Associate Professor**, Department of Computer Science and Engineering, Daffodil International University. We also declare that neither this projectnor any part of this project has been submitted elsewhere for the award of any degree or diploma.

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ABSTRACT

This project presents the development of an IoT-based smart energy meter utilizing an ESP32 microcontroller and the PZEM-004T energy meter module. The system measures real-time electrical parameters, including voltage, current, power, and energy consumption, and transmits the data over a Wi-Fi network. The energy data is then visualized through the ThingSpeak App, allowing remote monitoring of electrical usage. The integration of PZEM-004T with ESP32 enables accurate energy measurement and facilitates easy access to energy statistics via mobile devices. The project leverages the REST API & MQTT API to display the data in the form of channel fields on the app, providing users with intuitive feedback on their power consumption. This system offers an efficient solution for monitoring and managing energy consumption, contributing to energy conservation efforts and promoting awareness of power usage. The proposed energy meter can be further enhanced to include features like alerts, power optimization, and integration with other smart home systems.

turnitin Page 7 of 62 - AI Writing Submission Submission ID trn:oid:::1:3441647571

Table of Contents

Approval i

Declaration ii

Acknowledgements iii

Abstract iv

List of Figures vii

List of Tables viii

1 Introduction 1
- 1.1 Introduction 1
- 1.2 Motivation 1
- 1.3 Objectives 1
- 1.4 Methodology 2
- 1.5 Project Outcome 2
- 1.6 Organization of the Report 2

2 Background 3
- 2.1 Introduction 3
- 2.2 Literature Review 3
  - 2.2.1 Similar Applications 5
- 2.3 Gap Analysis 6
- 2.4 Summary 6

3 Research Methodology 7
- 3.1 Requirement Analysis & Design Specification 7
  - 3.1.1 Software Requirements 8
  - 3.1.2 Design Specification 9
  - 3.1.3 System Constraints 9
  - 3.1.4 Security Considerations 9
  - 3.1.5 Overview 10
  - 3.1.6 System Design 11
  - 3.1.7 Functional and Nonfunctional Requirements 12
- 3.2 Detailed Methodology and Design 12
- 3.3 Project Plan 17
- 3.4 Task Allocation 19

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3.5 Summary 19

4 Implementation and Results **20**
4.1 Environment Setup 20
4.2 Results and Discussion 21
4.3 Summary 23

5 Engineering Standards and Design Challenges **24**
5.1 Compliance with the Standards 24
5.1.1 Software Standards 24
5.1.2 Hardware Standards 27
5.1.3 Communication Standards 29
5.2 Impact on Society, Environment and Sustainability 32
5.2.1 Impact on Life 32
5.2.2 Impact on Society & Environment 34
5.2.3 Ethical Aspects 34
5.2.4 Sustainability Plan 35
5.3 Project Management and Financial Analysis 37
5.4 Complex Engineering Problem 41
5.4.1 Complex Problem Solving 41
5.4.2 Engineering Activities 44
5.5 Summary 46

6 Conclusion **47**
6.1 Summary 47
6.2 Limitation 47
6.3 Future Work 48

References **50**

List of Figures

3.1.1 ESP32 Microcontroller..... 7
3.1.2 PZEM-004T..... 8
3.1.3 Flowchart Diagram 11
3.2.1 Schematic Diagram..... 13
3.2.2 Circuit Diagram..... 14
3.2.3 User Interface..... 15
4.2.1 Final Result of our project..... 22

turnitin Page 10 of 62 - AI Writing Submission Submission ID trn:oid:::1:3441647571

List of Tables

2.1 Summary of Literature Reviewed. 3
2.2 Gap Analysis..... 6
5.1 Mapping with complex Engineering Problem..... 41
5.2 Mapping with knowledge Profile. 41
5.3 Mapping with complex engineering activities.....44

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Chapter 1

Introduction

This chapter Introduces the project, Motivation, Objectives, as well as the Methodology. It also discusses the Organization of the Report of the project and the Expected Outcomes.

1.1 Introduction

Due to the increasing demand of power and monitoring requirement, traditional energy meters fail in providing real-time reading along with remote access. The deficiencies of these systems are that the cost is high and data analysis or availability is not in real time, we propose here a smart energy meter with IoT using ESP32 microcontroller integrating PZEM-004T module helps real-time monitoring and remote access for viewing energy consumption details.

1.2 Motivation

With the growth in global energy usage and demand for better energy management, there has been a need for smart energy meters with real-time information coupled with analytics. By addressing this problem I hope to deliver a more affordable and impactful tool for monitoring energy usage that will enable people to make informed choices in order to cut down their consumption, lower bills and their impact on the environment. Furthermore, the possibility for remote energy consumption monitoring and controlling contributes to the emerging smart home concept as a convenient and energy-saving building device.

1.3 Objectives

The purpose of this project is to create an IoT energy meter with the ESP32 that you can monitor in real-time from your phone and has the capability to measure: power, current, voltage and calculate the energy consumption. Also, project's main goal is to enable accessing this energy data remotely via ThingSpeak app which will help in managing and optimizing the energy.

1.4 Methodology

The methodology in this work contains combining the ESP32 microcontroller and PZEM-004T energy meter for gathering real time information of energy consumption which consist voltage, current power and energy. The collected data is then sent over Wi-Fi to the ThingSpeak via the REST & MQTT API and visualized in channel fields for easy monitoring and analysis, enabling users to track their energy usage remotely.

1.5 Project Outcome

As a result of this work is an effective IOT based smart energy meter, it monitors the real time values of electricity consumption parameters like voltage, current power and energy. The system enables users to remotely monitor and track their energy consumption through the ThingSpeak app which encourages efficient power management, saving on energy cost and healthier lifestyle indices on sustainable use of energy.

1.6 Organization of the Report

Chapter 1: Introduction – Discusses the background, problem statement, motivation, objectives and methodology of the project.

Chapter 2: Literature Review – This chapter provides an overview of the state-of-the-art in smart energy meters and IoT enabled energy monitoring system

Chapter 3: System Design and Implementation – Describes the design, hardware list, software coding and integration of the implemented IoT-based power meter.

Chapter 4: Results and Discussion – This chapter discusses the results of the project with its data analysis, and evaluation of system performance.

Chapter 5: Engineering Standards – This chapter is about Compliance to the Standards, Effect on Society impact on Environment and sustainability, Project Mgt.& Financial Analysis. It also includes Complex Engineering Problems of this project.

Chapter 6: Conclusion and Future Work – Concludes about the achievements, discusses some limitations of the project, and provides recommendations on potential improvements and future enhancements.

Chapter 2

Background

This chapter introduces the Background, Literature Review & Gap analysis. It also discusses the Summary of this project.

2.1 Introduction

The explosive development of Internet of Things (IoT) has resulted in several interesting solutions for sensing and actuating many things (including energy consumption). However, traditional energy meters are only capable of measuring power consumptions and there no real-time data, remote accessibly and they are not smart integrated as new technologies. This has led to a need for smart energy meters that can deliver detailed information about power consumption as well as offer services such as remote monitoring, real-time data analytics and power-level optimization.

2.2 Literature Review

When I build up this project I need to do background study and read some literature. Here are those:

Table 2.1: Summary of Literature Reviewed.

Author (s)	Year	Title	Methodology	Key Findings
Kodali, R. K., et al.	2018	A Low-Cost Smart Energy Meter Using IoT	Used Arduino with ACS712 current sensor & voltage sensor. The data were transmitted to the ThingSpeak by ESP8266	Proven cost-effective design. Pointed out limitations to the accuracy of simple sensors compared with a single-metering chip.
Patel, H., & Patel, D.	2020	Real-Time Energy Monitoring Using ESP32 and IoT Cloud	Used ESP32 with SCT-013 current sensor. Data was sent to both Blynk and ThingSpeak.	Demonstrated that ESP32 was superior for integrated Wi-Fi. Dual platform (Blynk & ThingSpeak) data print SUCCESSFUL.
Shah, K., & Patel, M..	2019	IoT Energy Meter using PZEM-004T and NodeMCU with Blynk	Uses "PZEM-004T" with a interface module to adapt it to the micro-controller. Values sent to Blynk app by virtual pins.	Confirmed PZEM-004T provides high accuracy. Blynk works for easy mobile mornitoring in real time.

Sharma, P., & Gupta, S.	2021	Review on IoT-based Smart Energy Management Systems	Systematically review of 50+ research papers had been conducted regarding architectures, protocols and hardware.	ESP32/8266 are most popular MCUs. MQTT is the preferred protocol. There are dedicated chips (PZEM) which provide higher precision.
Shyam, G. K., et al.	2017	IoT Based Energy Meter Reading and Theft Detection	Worked on Arduino GSM module for data transmission through sms. Also added the most basic theft detection so far.	Constructed a model for energy theft detection, indicating that simple anomaly and basic logic can be used.
Shafi, Q., et al.	2018	Smart Metering: Challenges and Opportunities	A theoretical investigation of the concepts including network security, privacy, adoption..	Stressed that user trust, data protection and consumer benefits are as important to the problem as any technical solution.
Shahzad, G., et al.	2020	Home Energy Management System Using IoT and Cloud	Enabled Home Energy Management System Appliances controlled for on/off with schedule and tariff using microcontrollerwith relay. Used Node-RED for dashboard.	Extended supervision to active control for considerable cost cutting through automation and scheduling.
Sharma, N., & Mangla, M.	2019	IoT based Prepaid Energy Meter with Theft Detection	ESP8266 with energy metering circuit and relay. The system is power off when your account balance is low.	Demonstrated commercial potential for utilities using pre-paid systems and theft detection for minimized revenue loss.
Jaiswal, L. C., & Bankar, P. S.	2022	Real-Time Monitoring and Alerting System using Blynk	ESP32 was used along with the sensors. Set up Blynk events for sending email/push – on a high consumption etc.	Demonstrated how Blynk can be effective to deliver life-or-death user notifications in Email and push alert way. .
Muthuramalingam, S., et al.	2022	Comparative Study on IoT Protocols for Smart Energy	Experimental Comparison between MQTT, HTTP & CoAP in term of Power Consumption and Latency of ESP3237.	Determined that MQTT is the most network-efficient protocol because of it\'92s low power requirement and small data.

2.2.1 Similar Applications

Some smart energy monitoring systems using IoT technology have been deployed, which offered the same functionalities as we proposed in this system. Such applications are based on intelligent sensors, microcontrollers and communication devices for the real-time control of energy consumption. Some notable examples include:

1. **Smart Home Energy Monitoring Systems:** Companies like Sense and Neurio feature smart energy meters so homeowners can get real-time views of how power is being used to keep liters or kWh in check. These work by utilising Wi-Fi-enabled sensors based around the home, which send data to an app on a mobile device to allow users to view their electricity use as granularly as possible and spot power-hungry appliances.

2. **Smart Grid Systems:** At scale, smart energy meters based on IoT replace the traditional smart grid so that energy consumption gets optimized and waste minimized. These systems are employed by utility companies to track the flow of energy throughout the entire grid, giving consumers and suppliers access to real-time data so they can make better decisions about their energy use.

3. **Energy Management Systems (EMS):** EMS are commonly used by businesses and organizations to oversee energy consumption in several buildings. Systems such as Energy Star Portfolio Manager leverage IoT sensors to monitor energy consumption, which gives businesses visibility into areas where they can improve efficiency and lower costs.

4. **Blynk-based IoT Energy Monitors:** Apps created using Blynk are hot on home automation and energy monitoring features. We can access consumption data remotely using platforms such as Blynk through a smartphone app to track energy usage for checking trends, setting alerts on extreme usages.

These similar applications prove that there is a rising need (demand) and potential for IoT energy meters and the creation of an accessible real-time monitoring system which can be employed towards residential as well as commercial energy management.

2.3 Gap Analysis

Existing market has got many smart energy monitoring solutions, but, the few gaps that IoT based Smart Energy Meter system targets to provide are: • Cost reduction of existing energy monitoring systems. This difference suggests inadequacies by current solutions and further potential for enhancement; thus to bring new value added to users. The following are the main gaps in existing technology and how this project plans to resolve them:

Table 2.2: Gap Analysis.

Current System	Identified Gaps	IoT –Based Solution
No Real-Time Monitoring	Consumers cannot monitor their power usage in real time.	Real-time monitoring via mobile apps or web dashboards.
No Remote Access.	Users cannot access usage data.	Access to 'use of service' data and the ability to control usage.
Affordability issue	Dis-affordability for low class	Low cost-affordable for all classes
Expensive Setup & Integration	Consider additional costs for integration setup.	Ease of setup & integration
Only Final Usages	No Data Analytics & Insights.	Data Analytics & Insights
No Customization or Scalability	No user-define Customization	Provide the feature of Customization & Scalability
Complex User Interfaces	Hard to understand	User Friendly Interface & Experiences

2.4 Summary

In this chapter, we introduced the increasing requirements of IoT-connected smart meters (or energy metering devices), and the specific technologies which are employed to build such devices, like ESP32 microcontroller chip, PZEM-004T energy meter module. It presented similar usages of IoT for real time energy consumption monitoring with their advantages and limitations. A gap analysis was performed, finding that the inadequacies of current systems lie in affordability, remote enabling, setup simplicity and personalization. The aim of the proposed project is to fill this gap, by offering a low cost, user friendly and scalable system for real time energy monitoring that provides better flexibility, access and knowledge about the amount of energy consumed.

Chapter 3

Research Methodology

This chapter introduces the project Requirement Analysis & Design Specification, Detailed Methodology & Task Allocation. It also discusses Project Plan and Summery.

3.1 Analysis of Requirement & Design Specification

ESP32 Microcontroller: Serves as the master controller handling communication between the PZEM-004T module and ThingSpeak app.

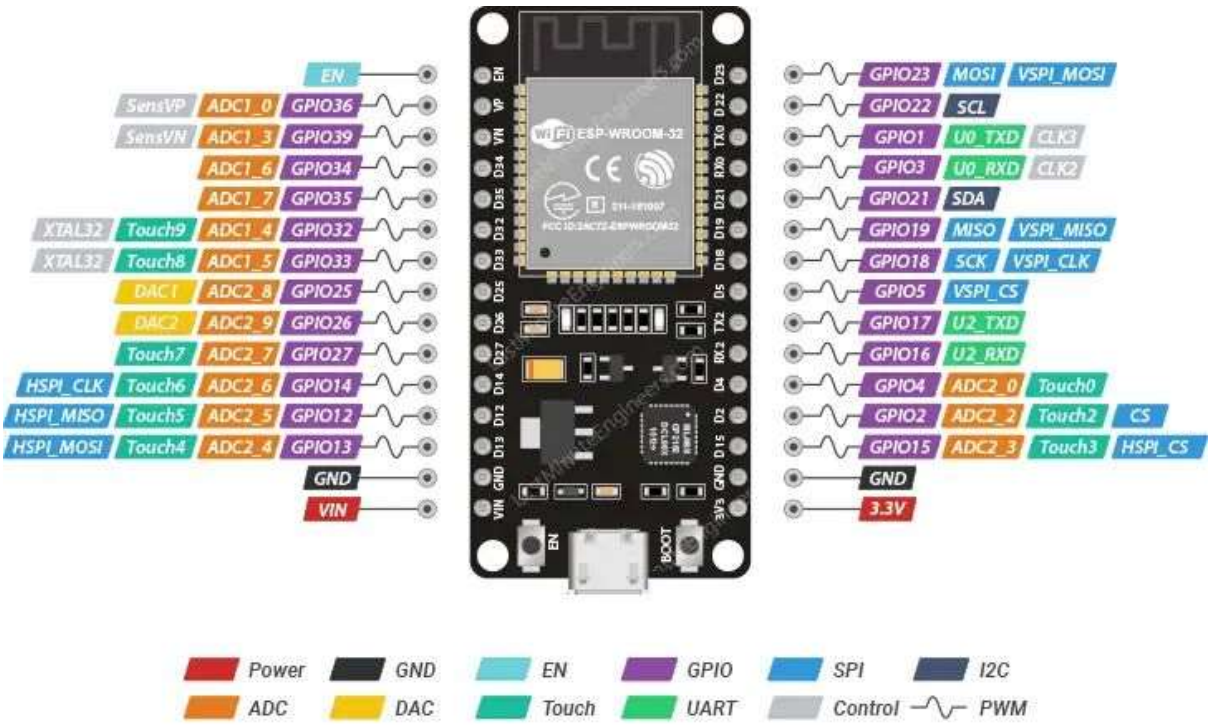


Figure 3.1.1: ESP32 Microcontroller

PZEM-004T Energy Meter Module: Can read voltage, current, power, and energy parameters.

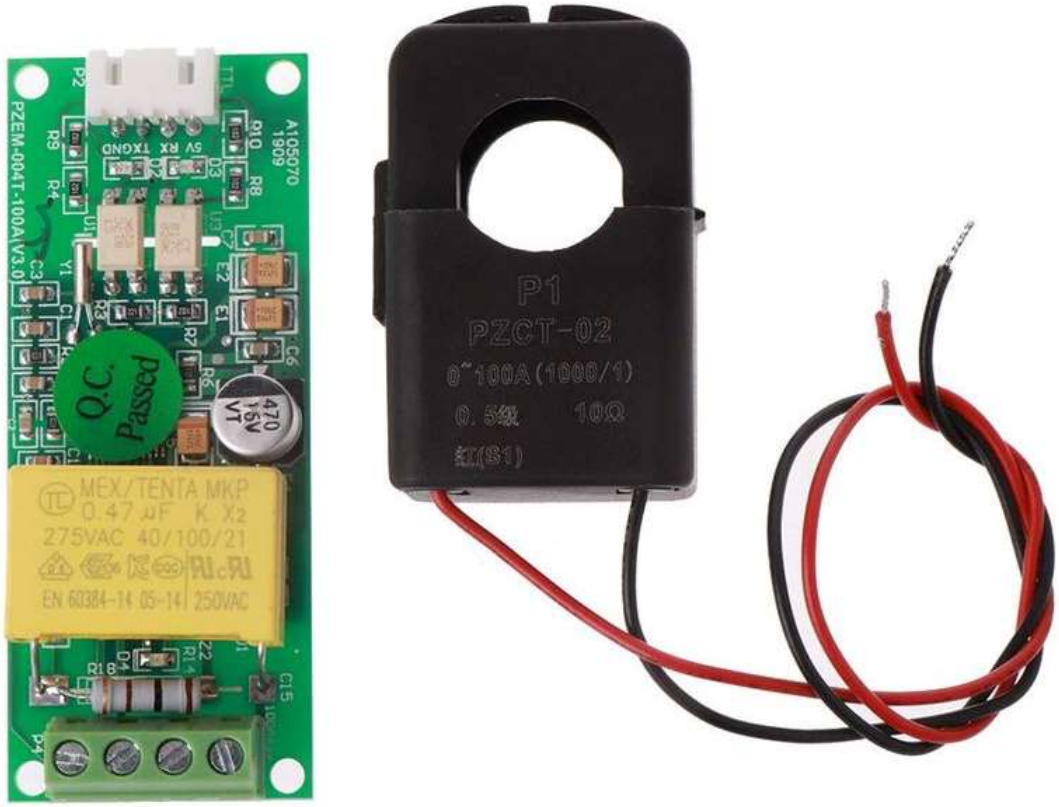


Figure 3.1.2: PZEM-004t

Power Supply: The device needs a power supply for the ESP32 and PZEM-004T.

Jumper Wires & Breadboard: For convenient hookup of components for prototyping.

3.1.1 Software Requirements

1. **Arduino IDE:** The tool we use to program the ESP32.
2. **ThingSpeak App:** The mobile interface will be developed for real-time monitoring and data visualization.
3. **ThingSpeak Library:** To let ESP32 communicate with Thing Speak app.
4. **PZEM-004T Library:** Let's make everything easy, the library is a communication protocol for PZEM-004T energy meter module.

3.1.2 Design Specification

1 **System Architecture:** The system include ESP32 with the PZEM-004T energy meter module, which take readings. This information is then transmitted over Wi-Fi to the ThingSpeak app for logging and viewing.

2 **Data Flow:**

- The PZEM-004T is used to measure energy and display it on the screen including Voltage, Current, Power and Energy.
- The ESP32 gathers data and sends it over Wi-Fi to the ThingSpeak application.
- The app ThingSpeak provides users with real-time data and the ability to track energy consumption from afar.

3 **UI/UX Design:** The interface of the application must have;

- Field the channel to display energy data on-the-fly (example: Voltage Field1,Current F2,Power F3,Frequency F4 & enery consumed so far in kWh is shown on field 5).
- Alarms and indicators of abnormalities in energy use.

3.1.3 System Constraints

1. **Power Consumption:** The unit itself must require only a small amount of power to provide reliable operation for 24/7 monitoring.
2. **Range of Operation:** It is important to consider the user's working environment where installation is planned as well The ESP32 can be connected to a network for data communication.
3. **Component Compatibility:** ESP32 and PZEM-004T should be compatible with each other, they have to work nicely without any problem in terms of data communication.

3.1.4 Security Considerations

1. **Data Security:** Data sent to the cloud or mobile app must be encrypted for privacy and security.
2. **User Authentication:** App should be accessible to only authorized users for monitoring and settings.

3.1.5 Overview

The proposed IoT based Smart Energy Meter is developed to measure and monitor the Electric consumption parameters like current, voltage, power and energy in real time basis. The system uses ESP32 microcontroller, and PZEM-004T energy meter module (display), that permits to get data remotely and in real time. Wi-Fi is used for data transmission to the cloud or a mobile app like ThingSpeak, where data can be easily visualized. The application is created to let the users have a friendly-use way in order to track their energy consumption and to understand how using or not using (saving) energy for a certain period. This allows users to maximise energy efficiency, save money and make smart decisions about how they use their energy. Additionally, the solution should be affordable, scalable and user-friendly enough for use in both home-based and small business applications. System cool points include real time monitoring, remote access to the data, notification customization and a simple data reporting interface through an app on your phone. This construction means our smart energy meter is more than just an effective tool to manage your energy, it's a cost-effective and dependable way for anyone who wants to reduce their electricity consumption.

3.1.6 System Design

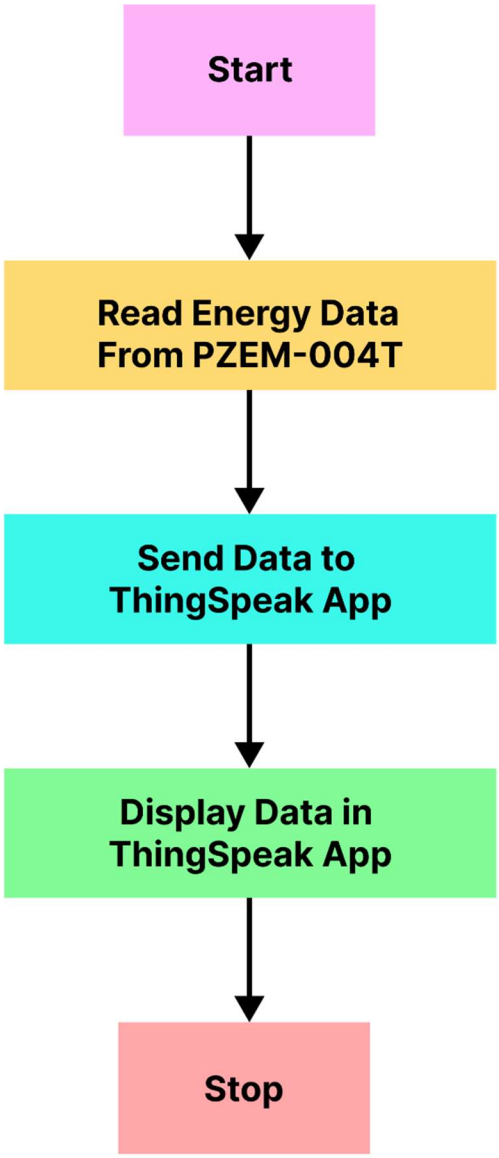


Figure 3.1.3: Flowchart diagram

turnitin Page 22 of 62 - AI Writing Submission Submission ID trn:oid:::1:3441647571

3.1.7 Functional and Nonfunctional Requirements

Functional Requirements

Real-Time Data Collection: The system should monitor energy parameters, such as voltage, current, power and energy consumption on per second basis.

Wireless Data Transmission: Send the data obtained to cloud service (e.g., ThingSpeak), computer, or mobile application.

Remote Monitoring: It should be to measure the energy on your Phone or a web based platform(online).

User Alerts: The system should be capable of user-defined sets levels and notify user if the value exceeds that level or to a condition previously defined by another user.

Data Visualization: There should be a visualization of energy consumption (voltage, current, power and energy) on the screen portrayed on mobile-app in real-time.

Non-Functional Requirements

Cost-Effectiveness: The system shall be cheap to acquire and utilize low-cost peripherals such as ESP32 and PZEM-004T.

Reliability: The solution should provide stable data delivery, and withstand downtime as well as real-time updates.

Scalability: The solution must be scalable to support expanding capability and deployment such as addition of new energy meters or advancement in analytics.

Ease of Use: The app must be easy to use and should have a user friendly interface in the mobile part for data display and setting up.

3.2 Detailed Methodology

The step by step process of the developed IoT-Based Smart Energy Meter is split into various activities such as hardware configuration, software strategy, data logging and integration with the ThingSpeak app. It is verified that the monitoring system can generate data and display energy situation in real time.

3.2.1 Hardware Setup

ESP32 Microcontroller: The ESP32 microcontroller is chosen as the central controller for the system due to its built-in Wi-Fi capabilities, allowing it to connect to the internet and communicate with the ThingSpeak app.

ESP32 Microcontroller: The ESP32 microcontroller will serve as the brain for this system because it has Wi-Fi features already installed that connect to the internet and with ThingSpeak app.

Connections: The ESP32 will be connected to the energy meter PZEM-004T using RX and TX (UART pins).

Power supply: A proper power supply is to be connected to ESP32 and PZEM-004T module. The ESP32 runs at 3.3V and the PZEM-004T is powered at 5V so correct voltage regulator or power supply should be used.

PZEM-004T Energy Meter: This tool will be used to calculate voltage, current, power and energy when operating. It then talks to ESP32 via serial communication (UART).

Data Acquisition: The PZEM-004T can monitor voltage (V), current (A), power (W) and energy consumption (Wh). A microcontroller ESP32 will periodically read the data.

WiFi Router: The ESP32 should connect to the WiFi router then we will be able to communicate with cloud sever ThingSpeak and send data to mobile app.

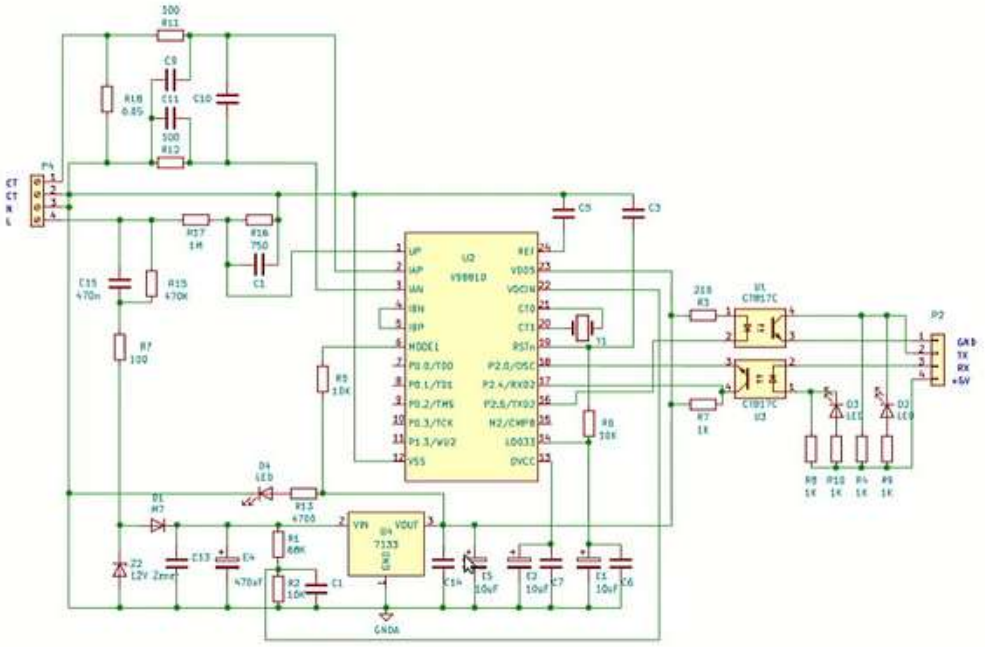


Figure 3.2.1: Schematic diagram

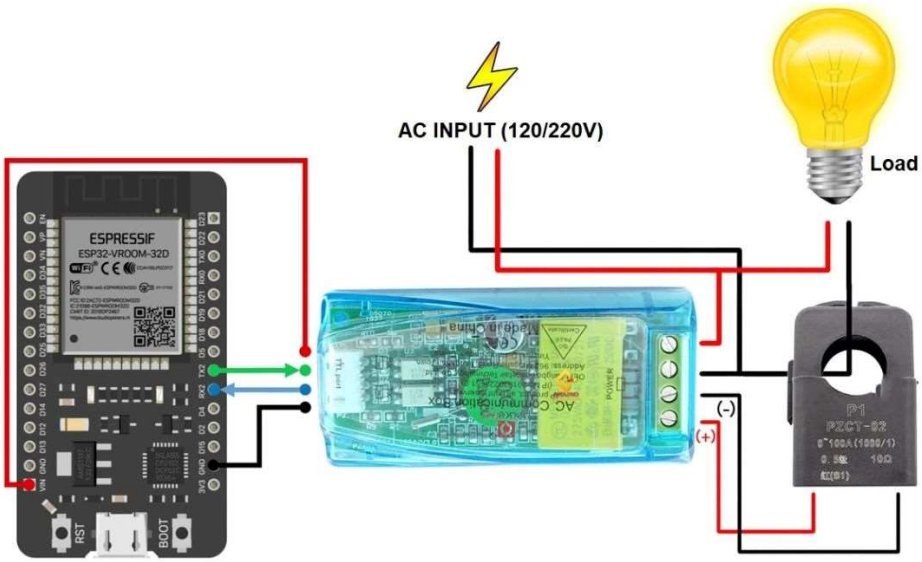


Figure 3.2.2: Circuit diagram

3.2.2 Software Development

Arduino IDE: ESP32 will be programmed using the Arduino IDE. The code will have functions that read data from the PZEM-004T, process and send it to the ThingSpeak app..

Libraries:

PZEM-004T Library: This library provides a simple way of communicating with the PZEM-004T energy monitor from ESP-IDF. Functions are present to read voltage, current, power and energy from the module.

ThingSpeak Library: This library creates a connection between the ESP32 and Thingspeak mobile app. It manages the data when transmitting over WiFi and sends real-time energy-usage information to the app.

1. **ThingSpeak App Configuration**:
    - The real time energy consumption data is received from ESP32 and the graphs are plotted accordingly for its visualization using ThingSpeak. Voltage, Current & Power Monitoring over Internet with ThingSpeak Channel.
    - Every value measured will be transferred to a field in the ThingSpeak channel:
        - Voltage → Field 1
        - Current → Field 2
        - Power → Field 3
        - Frequency → Field 4
        - Energy → Field 5

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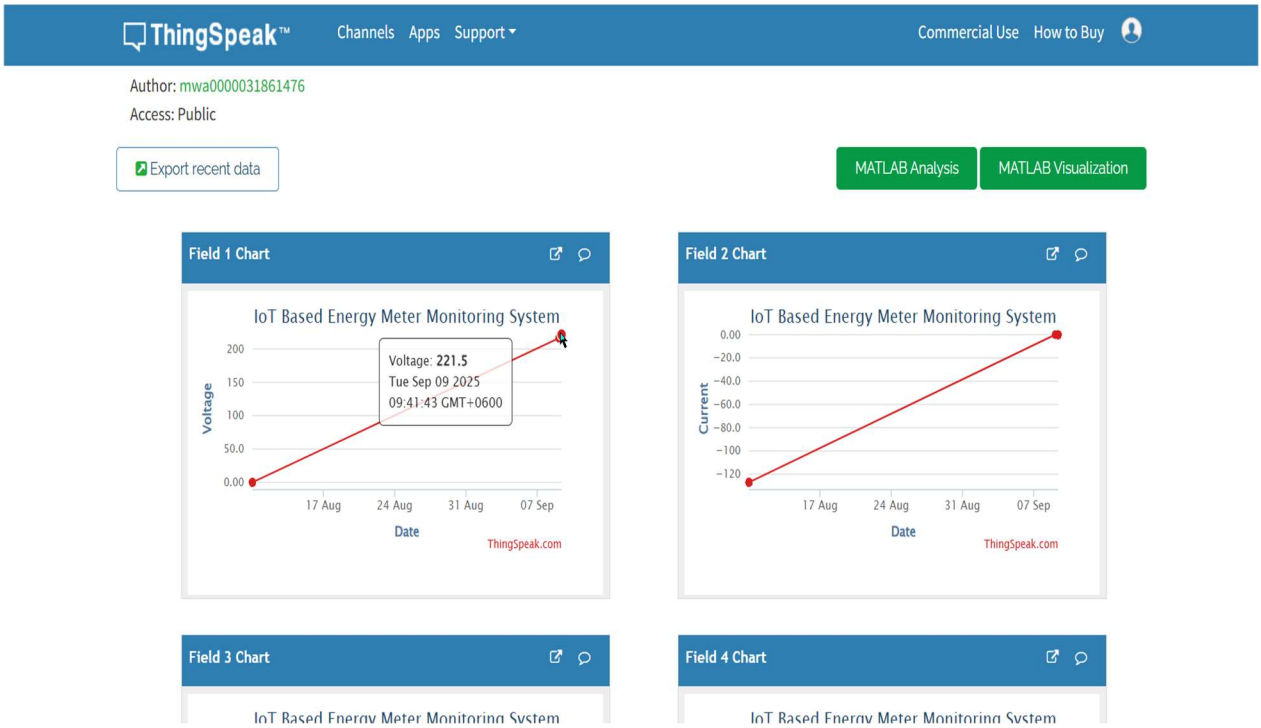


Figure 3.2.3: User Interface

3.2.3 Data Collection and Transmission

Reading Data: The ESP32 will be reading the data from the PZEM004-T at a time interval (e.g., every 10 seconds) by using some functions included within the library of the PZEM-004T. Voltage, current, power and energy data will be obtained from the PZEM-004T module.

1. **Data Transmission**: Once we have the data, the esp32 will connect to Wi-Fi and then send this data to ThingSpeak Cloud, from where mobile app will get that information. We will send the ESP32 data to ThingSpeak application using its WriteFields() function through channel Fields.
2. **Real-Time Updates**: The ThingSpeak app includes a view where mobile users can see the live data. Using graphs and live values, users can track their energy consumption and trends.
3. **User Interaction**: The application will enable users to configure alerts or thresholds. For instance, when the power consumption is higher than a predefined threshold, the system can push a notification to the user through ThingSpeak app.

3.2.4 System Integration and Testing

1. **Hardware Integration**: Hardware experiment by linking ESP32 and PZEM-

004T to acquire the right getting of data between them. The ESP32 will be on, I'll confirm it's obtaining readings from the PZEM-004T.

2. **Software Integration**: Upload the code to ESP32 Driver program using Arduino IDE. The software for the system will be tested by verifying that energy is being accurately sent to the ThingSpeak app.
3. **App Configuration**: We will test the ThingSpeak app to see whether all the channel fields are displaying the data, sent from ESP32 correctly.
4. **Final Testing**: The system will be thoroughly tested to verify that it operates in a realistic environment and energy data is transmitted flawlessly from the energy harvester to mobile app.

3.2.5 System Optimization

1. **Power Consumption**: The ESP32 power consumption will be designed to provide a continuous operation. Some sleep modes can be low power, such as when there is no active communication and overall power consumption can be lowered.
2. **Data Accuracy**: We will also test the validity of PZEM-004T readings, calibrating them if necessary and confirming that the device is sending us accurate information in the application ThingSpeak.
3. **Scalability**: The system will be designed to be scalable in the future, for using more energy meters for monitoring multiple devices or, for example, adding new features (e.g. advices on saving energy based on user's habits).

3.2.6 User Interface and Features

1. **Application Interface:** The ThingSpeak App will have a friendly interface to visualize energy consumption records. The app will be easy to use and easily navigable to show live readings, past data, trend analysis etc.
2. **Integration with the energy management app Push notifications:** Users will get alerted on their mobile when they consume more than a defined amount of energy.
3. **Data Analytics:** In the future, data analytics can be incorporated to enable users for making smart decisions with respect to optimizing energy usage based on his historical consumption information.

3.2.7 Security and Data Privacy

1. **Secure Encryption:** All communications between the ESP32 and the ThingSpeak app will be encrypted for security and privacy.
2. **User Authentication:** The ThingSpeak application will need a secure authentication facility placed on it, but the system as user log in / access against an username and passwords field.

turnitin Page 27 of 62 - AI Writing Submission Submission ID trn:oid:::1:3441647571

3.3 Project Plan

Phase 1: Research and Planning (5 Week)

Objective: To get the qualitative rooting and sign-off of the main project requirements.

Tasks:

- Review previous research on IoT energy monitoring systems.
- Find out which types of components (ESP32, PZEM-004T) and libraries that will be used.
- Define project goals and scope.
- Choice of connectivity: communication protocols (Wi-Fi and UART) and IoT platform (ThingSpeak).

Milestones:

- Finalized project scope and objectives.
- Identified hardware and software components.

Phase 2: Hardware Setup and Integration (3 Weeks)

Objective: To set up the hardware and verify communication between ESP32 and PZEM-004T.

Tasks:

- Put the ESP32 and PZEM-004T modules together.
- Configured UART communication between ESP32 & PZEM-004T to receive data.
- Turn on the devices and inspect the connection.

Milestones:

- Hardware successfully assembled and connected.
- Preliminary tests of the energy counts.

Phase 3: Software Development (5 Weeks)

Objective: Creation and implementation of software for data obtaining, transmitting and connection to the ThingSpeak app.

Tasks:

- Write program to read energy of PZEM-004T based on Arduino IDE.
- ThingSpeak library to add communication between ESP32 & Mobile app.
- Wi-Fi-based forwarding to cloud server ThingSpeak.
- Add a mobile app UI in ThingSpeak to show real-time data on channel fields.

turnitin Page 28 of 62 - AI Writing Submission Submission ID trn:oid:::1:3441647571

Milestones:

- Coding done and tested for energy meter reads.
- Fundamentals of communication set for ESP32 and ThingSpeak app.

Phase 4: System Testing and Debugging (3 Weeks)

Objective: Runnable a nd now upon sending the data to the system which stores it, with that the infamous Big Red Button should push us all into a new age baby! ; The idea was clear, check if from completing one end to another whilst pushing data trough our product does also show on ThingSpeak app.

Tasks:

- Real-time, online testing of the system accuracy (voltage, current, power, frequency & energy).
- Debug bugs related to communication or data processing.
- Stable Wi-Fi connectivity and data transfer between phone and Smartwatch to your device.
- Verify that the ThingSpeak App show data on button fields.

Milestones:

- Real-time, online testing of the system accuracy (voltage, current, power, frequency & energy).
- Debug bugs related to communication or data processing.
- Stable Wi-Fi connectivity and data transfer between phone and Smartwatch to your device.
- Verify that the ThingSpeak App show data on button fields

Phase 5: User Interface Design and Final Testing (3 Week)

Objective: Finish the user-interface design and final testing to ensure the usefulness of stability.

Tasks:

- Update the ThingSpeak app UI to make it more intuitive and easier to follow.
- Perform final tests throughout the system.
- Alert users when certain power consumption levels are exceeded with push notifications or alerts.

Milestones:

- User interface finalized and refined.
- System tested under real-world conditions.

turnitin Page 29 of 62 - AI Writing Submission Submission ID trn:oid:::1:3441647571

Phase 6: Deployment and Documentation (3 Week)

Objective: To release the system for use and to produce sufficient documentation.

Tasks:

- Demonstrate the smart energy meter in live performance in actual field setup.
- Issue user documentation for system installation and use.
- Write report containing findings, limitations and recommendation for future work.

Milestones:

- Deploy with the system up and running in a real environment.
- Documentation prepared and finalized.

3.4 Task Allocation

A timeline of the major project events in each iteration is shown in this table from week 12 to week 48.

Tasks	Weeks																			
	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48
Research and Planning																			
Hardware Setup and Integration																			
Software Development																			
System Testing and Debugging																			
UI Design and Final Testing																			
Deployment & Documentation																			

3.5 Summary

In this section, we describe the task allocation for our IoT-based Smart Energy Meter project which has been divided into six main phases: Research and Planning, Hardware Setup and Integration, Software Development, System Testing and Debugging, User Interface Design and Final Testing and Deployment and Documentation. Tasks are

divided among different members like – project manager, hardware engineers, developers, mobile app designers, testers and documentation people. Roles are clearly defined for each member, so that the project is efficiently completed and successful delivered though working together effectively using skills prudently to stay on schedule.

turnitin Page 31 of 62 - AI Writing Submission Submission ID trn:oid:::1:3441647571

Chapter 4

Implementation and Results

This chapter introduces the project Environment Setup. It also discusses Results and Discussion of this project.

4.1 Environment Setup

Basic Functionality:

Once you have everything setup, and working on the hardware and software side, load onto your ESP32 a basic test program to verify that it can communicate with the PZEM-004T energy meter as well as the ThingSpeak app. Ensure the energy data (voltage, current, power, energy) value is getting accurate from PZEM-004T and display on ThingSpeakApp.

Wi-Fi Connection:

Verify that the Wi-Fi connection works between the ESP32 and the Wi-Fi router (you should see: connected) as well as send data to ThingSpeak cloud server.

Error Checking:

Don't wait for system integration testing to find out the hardware or software is not working.

turnitin Page 32 of 62 - AI Writing Submission Submission ID trn:oid:::1:3441647571

4.2 Results and Discussion

In this section, the results obtained from implementation of the IoT-based Smart Energy Meter system are showed and its discussion towards objectives, performance of system, real world scenario has been made. This will include data analysis, system test results, and any observations of the project's evaluation.



Figure 4.2.1: Final Result of our project

4.2.1 System Performance

The smart energy meter was efficiently connected and verified that the ESP32, PZEM-004T and ThingSpeak App were operating normally. The energy consumption data (voltage, current, power and energy) was measured and sent by the system over a Wi-Fi network. The system operated as follows:

Data Feasibility: It means that the system can display voltage, current, power and energy consumpion in nearly real time. Output of the PZEM-004T module compared to a

standard energy meter, readings verified for accuracy. The measurements from the two systems were not significantly different indicating that there was no bias and therefore demonstrating that good data for monitoring was provided by the smart energy meter.

Data Transmission to ThingSpeak: ESP32 successfully transmitted data using Wi-Fi to the ThingSpeak app. The information was shown to the user through channel field, live and near real-time so there could be a direct feedback to energy consumption. Data loss on the data transmitting process did not occur, and an constant performance was provided while it is tested.

Monitoring: Users could check real-time energy consumption remotely from ThingSpeak app. The app was intuitive and offered real-time updates so users could interpret their power consumption trends in realtime. This was particularly useful to spot out greedy devices and reduce power consumption.

4.3 Summary

The IoT-Level Smart Energy Meter was designed and developed, such that it can be efficiently installed and operated. We were able to establish the Smart Energy Meter system on an experimental basis to achieve the short-term objectives of real-time energy measurement capability, remote transmission control based web interface with user interaction functionally. The terminal was capable of accurately measuring voltage, current, power and energy of consumption, uploading the real-time data fed back to ThingSpeak application for remote monitoring. Although it was smoothly integrated and worked well altogether, a couple of drawbacks were revealed, that is the need of stable Wi-Fi connection and the power consumption from all its components. But we're hoping that these issues can be resolved in later iterations - both the range of the Wi-Fi, and optimizing power to work continuously. Users reported ease of use and the convenience of tracking energy consumption remotely was specifically described as popular. The picture has been suggested to users that they wanted net, supporting features such as historical data analysis and offer energy-saving tips supported. Compared to the commercial solutions, this project provided a cheaper solution that can be expanded and completely customized. In general, the system proved to be able to encourage energy saving and informed control over power usage. Further development could consider improvements in the security, scalability for larger setups, and also integration with other smart home technologies to enable a more complete home energy management.

Chapter 5

Engineering Standards and Design Challenges

This chapter describe Compliance with the Standards, Impact on Society, Environment and Sustainability, Project Management and Financial Analysis. It also discusses Complex Engineering Problems of this project.

5.1 Compliance with the Standards

Considerable attention was put in the construction of IoT Smart Energy Meter system to guarantee the compliance with industry standards, safety norms and best practices. International Standards This product components of PZEM-004T with the ESP32 micro computer certified to meet the relevant international safety standards (IEC 61010-1) and compliance with EMC energy meter, to prevent electrical shock hazard and overcurrent condition. The system is also CE certified, which demonstrates that it meets safety, health and environmental protection standards in the European Economic Area. In terms of connectivity, communication protocols were established through an IEEE 802.11 Wi-Fi standard for data transfer and ICP/IPC connection to provide compatibility with the most networks. Moreover, secure TLS/SSL encrypted communication guarantees that exchanged data remains confidential between provider and consumer. The design complies with environmental regulations such as RoHS meaning that hazardous materials are not used in the components and power consumption is kept at a minimum with input low-power modes. Additionally, we consider issues concerning data security by following privacy regulations such as the EU General Data Protection Regulation (GDPR) to guarantee secure management of user information. What's more, it was designed with industry-standard best practices for IoT, such as scalable modular architecture and a user-centric mobile app experience. As the project moves towards commercialization there will be an increasing requirement for future regulatory compliance with certifications such as FCC and national energy meter standards. Those measures ensure that the system is reliable, secure and green enough to meet legal and industrial demands which are required.

5.1.1 Software Standards

It is essential that while developing the IoT-based SEM system, certain software standards and best practices were followed to guarantee system reliability, security, scalability. Below is an overview of the most important SW standards and guidelines that were applied during development:

5.1.1.1 Code Quality and Maintainability

Good coding practices and industry best standards followed during the software development were:

Modular Programming: The code was written in the form of modular blocks so that each feature such as reading data from PZEM-004T and transmitting it to ThingSpeak app have been encapsulated in different modules. This modular design will facilitate the subsequent maintenance, debugging and future extension.

Commenting: Effective commenting was implemented throughout the coding process,every function and significant code block was annotated. This will benefit other devs coming later up the road, and ease them into maintaining things in the future.

Version Control: Git served as version control for the codebase handling changes in code. This not only makes the developing process be able to see clearly but also can quickly locate and prevent code conflicting, as well as enhance the development of team cooperation!

5.1.1.2 Programming Standards

The program was written in the Arduino IDE (for ESP32 microcontroller), according to the programming instructions below:

Library Compatibility: We used commonly used and well-maintained libraries like PZEM-004T library, ThingSpeak library that allowed assure compatibility as well as ease of use for integrating energy data from ESP into the apps on ThingSpeak.cloud. These libraries are frequently updated to provide better features and security.

Error Handling: The solution has appropriate error handling and exception management strategy to deal with issues faced during data collection, transmission and ThingSpeak app interaction. As long as the system is able to deal nicely with errors, it can continue to work stably and effectively under different working environments.

Cross-Platform Compatibility: Code was written in a cross-platform way to be compatible across various operating systems such that replication of both the development and deployment environment is simple. And this is very important for test and debug.

5.1.1.3 Security Standards

Because of the nature of the system, security was paramount in its construction:

Data Encryption: The system complies with secure communication standards – the information transmitted from the ESP32 to ThingSpeak cloud server is encrypted using TLS/SSL. This encrypted process keeps energy consumption information sensitive and safe during transmission, deterring unauthorized access or tampering.

Authentication: The ThingSpeak app used secure authentication methods to manage unauthorized access. It is user privacy as well as data protection Users can only view their power consumption if registered.

Secure Coding Practices: The application was developed in accordance with secure coding principles, such as not hardcoding sensitive information(passwords, API keys), using best practices for input validation and output sanitization so as to prevent vulnerabilities.

5.1.1.4 Adherent to IoT Design Guidelines

The software was designed based on certain IoT specific SW development standards, allowing achieving interoperability, scalability and an effective integration with other IoT devices. These include:

Interoperability: The system is developed based on standard protocols like MQTT (Message Queuing Telemetry Transport) for message transfer among the devices, which facilitates interoperability between different IoT systems and cloud platforms.

Energy Efficient: The software incorporates power-saving functions to enter sleep mode during inactivity, making the IoT system energy efficient and viable for constant long-term operation.

Scalable: The program framework is designed to accommodate features added in the future as well as externals devices or capabilities without having to redesign the system. This flexibility is important as it will allow the solution to grow to meet future requirements (eg supporting more energy meters, other smart home systems etc.)

5.1.1.5 User Interface Standards

The ThingSpeak application front-end conformed with general principles of mobile app design to ensure an easy-to-use interface:

Consistency: The application design follows the UI/UX conventions providing a level of coherence in navigation, screen and data presentation. This uniformity increases the usability of the system and ensures that it can be easily used by users of all skill levels.

Responsiveness: The app is responsive and there were minimal to no late updates. It is designed for mobile devices, so that users can interact with the system and visualize their energy usage while they are on the move.

5.1.2 Hardware Standards

The Hardware components chosen for IoT-based Smart Energy Meter system are both standard compliant and adhere to industry best practices that guarantees safe, reliable and efficient operation. The main hardware used, i.e., ESP32 microcontroller2, PZEM-004T energy meter module3 and other related circuitry are compliant with existing system compatibility to ensure correct functionality and integration. Below are the major hardware standards.

5.1.2.1 Electrical Safety Standards

To guarantee that the system operates safely, the hardware units also conform to international electrical standards for electric and electronic devices:

IEC 61010-1 (Safety Requirements for Electrical Equipment for Measurement, Control, and Laboratory Use): Components like PZEM-004T power meter complying to the IEC 61010-1 ensures equipment is designed to protect against electrical shock and minimize risk of injury from normal operating hazards.

CE Certification: ESP32 and PZEM-004T modules are certified by CE (Conformité Européene), guaranteeing safety of these products and that they have a minimum impact on the environment in the European Economic Area (EEA). This certification is vital in guaranteeing that the parts when used under operation does not cause any harm to health, safety, and environment.

UL Approved: All components in the system are UL (Underwriters Laboratories) listed ensuring electrical components meet safety standards to be used in different regions such as North America.

5.1.2.2 Component Compatibility

The hardware system was made compatible to enable components (i.e., ESP32 microcontroller and PZEM-004T energy meter module) interoperability:

Electrical Interface Matching: ESP32 & PZEM-004T Serial COM port: The communication of electro-optical signal transmits based on UART interface and ensures the compatibility of measuring meter and other products. To keep quality transmission and system stability, the voltage of each signal sent is adjusted from one part of the module to another.

Voltage regulation and power supply standardization: The ESP32 is 3.3V, but PZEM-004T is 5V with a voltage regulator that guarantees safe operation of all the components. System conforms to power supply standards to protect against sudden voltage surges or drops which can cause this system to malfunction.

5.1.2.3 Environmental Standards

Following are the hardware compliance standards for the system, as specified in the respective sections: For better environmental protection, these components have met or exceeded some of those standards.

RoHS Certified The PZEM-004T and ESP32 modules do not contain environmentally harmful substances, depending on application lead, mercury, etc., making it an ideal design solution for transmission of data at high speed. This renders the components environmentally friendly and disposable in a manner that does not adversely impact the environment.

WEEE Compliance (Waste Electrical and Electronic Equipment): The components are designed to be compliant with WEEE, which allows them to be recyclable and disassembled at end-of-life.

Energy Saving: This system was designed to energy save. The ESP32 device is a power saving machine which can enter the hibernation state when not in use, consuming very little current to maintain the performance of all peripherals. This adds to the net efficiency of the system.

5.1.2.4 Mechanical and Design Standards

The hardware components are mechanically and aesthetically designed to standard specifications that guarantee durability, ease of mounting and safe use:

Compact and modular design: The hardware structure is designed to be compact and modular, which facilitates maintenance. The ESP32 and PZEM-004T are connected to breadboard or custom PCB for testing or modification.

Cons Thermal Solution: The components are chosen to run within their ideal temperature ranges. Both the PZEM-004T and ESP32 are in industrial/military grade, in this way we can use them even on harsh ambient. The design factors in heat dissipation to avoid overheating when using extensively.

Standardized Connectors: The manufacturers standard, UART communication connection port and power input and WIFI connectors all ensure that your-in-flight FPV feed stays as rock solid steady as possible. The advantage is that there will be no risk of loose connection or wrong wiring when installed and used up.

5.1.2.5 Wireless Communication Standards

The microcontroller is a O*F PCM5102A ESP32, which complies with the following protocols over Wi-Fi:

IEEE 802.11b/g/nYes (Wi-Fi standards supported by ESP32): The ESP32 supports 802.11

b/g/n (2.4 GHz) Wi-Fi No more false rumours that 5G IoT is "just around the corner". The standards can be used for an optimal communication between the microcontroller and ThingSpeak cloud based on real time monitoring of energy through mobile application.

Wi-Fi security standards (WPA2): The ESP32 utilizes WPA2, the latest Wi-Fi security standard for authentication and encryption of wireless connections between the system and the Wi-Fi network. Data transmission is protected against eavesdropping using a secure link with 128-bit AES encryption.

5.1.2.6 System Testing and Compliance

All hardware devices were tested in order to meet the applicable standards and functioning properly:

Functional tests: The functionality of the system was tested with ESP32 and PZEM-004T properly communicating and providing accurate measurements for energy consumption. Reliability and performance of the hardware and software was confirmed.

Safety Check: This power supply full course examination to ensure the safety, all electrical performance comply with design standards for protection of short circuit/ overload/ over voltage/ over current/ over temperature.

5.1.3 Communication Standards

The communication among the devices, i.e., ESP32 micro-controller, PZEM-004T energy meter and ThingSpeak app in the IoT based Smart Energy Meter follows some standard communication standards. These guidelines guarantee that the data communication of the system is effective, accurate and safe. Here are the communication protocols we follow in the project:

5.1.3.1 Wi-Fi Communication Standards

The major communication approach is incorporated around the Wi-Fi via the cloud to enable wireless transmission of data from ESP32 to ThingSpeak application. WiFi communication standards were followed as subsequent:

IEEE 802.11b/g/n (Wi-Fi, WiFi, WLAN): The ESP32 gives you the IEEE 802.11b/g/n standard Wi-Fi as well as Bluetooth v4.2 BR/EDR and BLE. These are so commonly used you will find compatibility with most home and business WiFi networks. This enables the system to transmit on Wi-Fi and provide stable, rapid data transmission for real-time energy monitoring.

Wi-Fi Security Standard (WPA2): WPA2 encryption for communicating securely over Wi-Fi provides standard security. This way the data sent end-to-end from ESP32 to ThingSpeak cloud server is secured from unauthorized access. The encryption scheme

contributes to the integrity and confidentially of transmitted data, which is important for IoT applications like the energy consumption measurement.

5.1.3.2 Serial Communication Standards (UART)

The ESP32 is communicated with the PZEM-004T energy meter module using the Universal Asynchronous Receiver-Transmitter (UART) protocol. The following standards were followed:

RS-232: (Recommended Standard 232) PZEM-004T has UART for interfacing but uses RS-232 signaling voltage levels ans this standard as a whole specification only the signals are represented in our uses. This protocol is generally used in embedded applications for sending command and control information to a device over a data channel, offering a reliable, asynchronous way of communication between an ESP32 microcontroller and the energy meter.

Data Integrity: The ESP32 and PZEM-004T communicate with mechanism to keep the data intact by error checks while transmitting. Furthermore, if the system detects any discrepancies in the data, it re-requests for that information to be read from the energy meter so that accurate information may be communicated.

5.1.3.3 MQTT Protocol (Optional for Future Expansion)

Later releases support MQTT (Message Queuing Telemetry Transport) protocol for lightweight, efficient messaging by the ESP32 to cloud services such as ThingSpeak. MQTT is also a highly suitable protocol in IoT systems due to its consumption of minimal bandwidth and support of intermittent connections. The following are prominent MQTT standards in this project:

Publish/Subscribe Model: MQTT uses a publish/subscribe model in which ESP32 publishes the data (energy readings) to the cloud server and ThingSpeak app subscribes for the updates. This model is cost-effective and reduces network consumption since data are sent only when there are changes.

Quality of Service (QoS): MQTT provides several QoS levels so that messages can be delivered at the desirable reliability. In this paradigm, QoS level 1 (at least once delivery) is acceptable to guarantee a reliable energy data transmission to the ThingSpeak app.

5.1.3.4 Cloud Communication Standards

The communication from ESP32 to ThingSpeak cloud server is using the HTTP/HTTPS protocol and is secure and efficient. This is a must have for ThingSpeak app real-time data monitoring & visualization. The standards followed are:

HTTP/HTTPS (HyperText Transfer Protocol / Secure HTTP): The ESP32 utilizes HTTP/HTTPS for data forwarding towards the ThingSpeak cloud server. HTTPS secures the transmitted data by encrypting, which prevents man-in-the-middle attacks and snooping.

REST API (Application Programming Interface): The ThingSpeak is equipped with REST API for the integration of IoT devices. The ThingSpeak cloud server is contacted by the ESP32 using this API and here, energy data through channel field can be input to the ThingSpeak app's real-time.

5.1.3.5 Data Encryption Standards

Security Measures To safeguard the trustworthiness and confidentiality of the information, the authors adheres to data encryption protocols:

Encryption (Wi-Fi and Cloud Compatible): The system encrypts data using Transport Layer Security (TLS), which is derived from Secure Sockets Layer (SSL), while transmitting the encrypted information between entering and leaving points; including almost always Wi-Fi at this time. This encryption keeps energy consumption data safe from unauthorized access, securing against messages being sniffed between the ESP32, ThingSpeak cloud server and mobile app.

End-to-End Encryption: Data is encrypted during all dispositions (from ESP32 to cloud side and at the ThingSpeak app's end), so the whole communication channel is actually secure.

5.1.3.6 Interoperability Standards

To endure the interconnection, with other IoT devices/enviroments in the future, high level communication architecture complies with an interoperability standard:

JSON (JavaScript Object Notation): The ESP32 and ThingSpeak cloud server communicate data in text-based JSON format, which is a lightweight data- interchange format that is easy for humans to read and write, as well as easy for machines to parse and generate. JSON makes it so the data can be parsed and understood by other devices or systems, which may integrate with the system at some point down the line.

Open API: ThingSpeak is an open-aspect platform, that can potentially be integrated with other smart platforms and devices in future. By following these open standards, the system is as flexible and scalable enough to add more devices or scale the application if necessary.

5.2 Impact on Society, Environment and Sustainability

The Proposed work on IoT based Smart Energy Meter Solutions has great positive impact on society, environment and sustainability. It allows for real-time energy consumption monitoring help users take better decisions on their energy usage leading to sustainable and efficient employers in residential, commercial and industrial areas. Below we provide an in-depth look at the systems impact on these three areas:

5.2.1 Impact on Life

Smart Plug and Phone: (Smart Energy Meter IoT System) Smart Energy Meter with the help of different techniques are far reaching in day to day life, which improve human's daily living by interacting individuals and industry with their usage control energy consumption. With this, as well as being fast and efficient it also has a positive impact on how people live by making them make smarter energy consumption choices. Here are a few ways the system affects life:

1. **Giving Consumers Control and Awareness**

The product allows users to own their energy and monitor real time data easily. Because people can see the patterns of their energy use on the ThingSpeak app, they become more conscious of how much and when they are using electricity. This consciousness allows users to visualize how and where they can cut down on waste, which in turn results in more efficient energy consumption. The behavioral change of people over time to turn off devices at stand-by position or use it smartly (during the non-peek hours) inherently saves money and minimizes impact on environment.

2. **Cost reductions and financial advantages**

Real-time tracking of energy usage has real economic value for the user. Since the current usage of electric can be monitored in real time, users can make timely decisions to save energy, this would result in lower electricity bills. In homes, especially higher consumption homes, this can lead to great cost savings. Small companies that implement the system can also save on operating costs based on where energy is being wasted and adjusting accordingly. In high energy cost areas, such an economy can go a long way to working yourself out of financial trouble.

3. **Enhanced quality of life through ENERGY CONSERVATION**

It promotes environmentally-friendly lifestyles by promoting energy saving. By way of illustration, through cutting energy waste the system prevents households and companies being lumbered with any unnecessary heavy electricity bills and this means people can

use it to pay for other things. In addition, energy efficiency contributes to a more comfortable home environment in that HVAC systems, lighting and appliances perform efficiently without wasted heat or cooling.

4 Improved Health and Comfort

Better energy management isn't just money and sustainability—it's also better health and comfort. Using energy efficiently means that energy consuming services like air conditioning or heating systems work effectively to maintain a comfortable environment but are not used excessively. The smart system can also be linked to your smart home devices for automatic real-time adjustments. For example, one could set temperature preferences or maximizes the appliance usage for better air quality and comfort, an action which directly have an impact on the QoL in a household.

5 Intelligent technologies for the mass audience

The low cost of the system could enable smart energy technology to be deployed to a wider section of the public, including those on lower incomes that may not otherwise have access to such technology. This inclusive approach also supports to close the digital divide and gives access to everyone in order to benefit energy-efficient deomination actions. The system makes smart tech accessible to everyone – making it easier for more people to engage in the greater sustainability movement and take advantage of the benefits modern IoT systems offer.

6 Supporting Well-being of Community and Environment

Add up the contributions of a lot of people or companies, and energy-saving efforts can add up to big benefits for communities. When people are more aware of their power use, total electricity demand can drop — which means both lower utility bills for individuals and relief on local power grids. Decreasing the demand for energy to move power plants into production can also help keep black outs from happening as well as minimizing pollution in the air while promoting a cleaner, more healthful community environment.

7 Promoting Sustainable Lifestyles

Through delivery of actionable information and enabling users to make informed choices, the system translates into more sustainable lives. People are more likely to be green if they can see the direct benefit of their measures in terms of both bills, and the environment. By documenting power usage in real time, the system encourages users to reduce wasteful energy usage, incentivizes them to install renewable sources like solar panels and motivates overall greater efforts at conservation – all of which add up to a more mindful way of living.

5.2.2 Impact on Society & Environment

The system of IoT based Smart Energy Meter has the great influence on social and environmental tools like enhancing efficient energy use, awareness raising, and into practice. Societally, the platform enables people and businesses to manage their own energy usage – offering live feedback on how much they are using. This deeper understanding drives action, such as shutting off idle electronics or shifting energy use to a non-peak time of day—and in turn realizes financial savings. This results in paying less for electricity, and is good for the household budget of lower bills. Moreover, the price point and relative ease of access to our full platform ensure that smart energy technology is available on a wider scale to more people, including those in lower-income homes, which closes the technological divide by providing more people with access to energy-saving solutions. As smart technology becomes more common, the vision of Smart Cities and Communities begins to emerge, with collective best utility from energy usage.

From the environmental standpoint, however, the system also serves to alleviate waste of energy by providing such users with information which helps them identify inefficient use of energy and take-action in relation thereto. This results in a decrease in total energy requirement, and consequently reduces dependence on nonrenewable sources for generation of energy, thus lowering the carbon footprint. In addition, the painted area panel of 135 sq m and double guillotine design mean that horizontal axis turbines can be integrated thus enabling users to match consumption with when renewable energy is generated ensuring a cleaner carbon footprint. Through promoting power saving and encouraging the end users to embrace sustainable behaviors, it also helps achieve wider global sustainability objectives such as reducing carbon footprint and natural resources conservation. The IoT based Smart Energy Meter system's effects are significant when it comes to promoting environment-friendly and energy efficient activities — good for the Society and Planet both.

5.2.3 Ethical Aspects

However, the IoT-enabled SEM system is faced with numerous ethical issues which need to be tackled so that it can develop and operate in an appropriate way. Privacy of information and data security is one of the major concerns. The system could enable real-time energy consumption monitoring, potentially exposing sensitive information about users' daily routines. Second, people need to know what data is being collected and for what purpose and be given clear choices about controlling or deleting that data. A third

key ethical consideration is security. The IoT systems can be targeted by hackers and successful attacks can result in unauthorized access, secure authentication techniques as well as up-to-date security measures should be used to prevent intrusions and protect user data. Access and democratization are other significant ethical considerations. We must build an Internet that is for everybody, regardless of their technical know-how—or bank balance. This involves developing a user-friendly interface and making the system accessible to diverse users. In addition, the system should be environmentally friendly and used sustainable materials for energy optimization. Candor and accountability are essential elements in preserving trust. Users need to be informed in a transparent way about how everything works and which data is collected and used. Transparency regarding 1) privacy notices and 2) terms of service should be adopted to enable users to have a full understanding of their rights. We should also consider at the end, the long-term social consequences of such system to be sure that it provides some added-value instead exploitation of a vulnerable group. The aim should be to ensure that energy-efficient technology is available to everyone. By tackling them, the system can provide further technological progress and benefit to society.

5.2.4 Sustainability Plan

Sustainability of the IoT-Smart Energy Meter system the sustainability plan meets the assurance that considers maintenance, political changes and how development will be sustained to guarantee a long time survival and influence [19]. The system is based on the foundation of energy efficiency, efficient use of resources and availability. On an environmental standpoint, the invention helps save energy since users can monitor and better manage their consumption of energy, which resulting in a decreased need for energy and a reduction in artificial carbon footprint. The features of reusable modules such as ESP32 micro-controller and PZEM-004T energy meter, they are all environmental-friendly to be RoHS complaint without toxic material. Furthermore, the system promotes adoption of renewable energy, enabling users to match their electricity consumption with available clean generation and further reduce its environmental impact.

On the social side, the system is intended for a broad user group and even low-income households are provided with reach to the system so those groups don't need to feel excluded. The system is also centered on learning and awareness-building by providing the end-user with the information necessary to embrace more sustainable use of energy. Community engagement is also another factor, sharing energy saving tips with each other

can inspire others and make everyone to do everything they can for better energy management. Technologically, the machinery is scalable and also future-proofed, so new developments can be added to or incorporated into existing smart home or energy management systems. This guarantees the system to stay flexible with changing energy demands and new technologies.

Ultimately, PERMAN lucre operative sustainably for the world as a whole by going to other places and entering initiatives agreements with governments/Organizations on how to make energy using technologies what is most appropriate there. By correlating with the United Nations' Sustainable Development Goals on affordable clean energy and climate action, in particular, the system aims at actively contributing to global environmental issues. The sustainability plan focuses not only on the system's green qualities, but also, it's ability to make a long-term social and environmental impact by changing the way people, communities and businesses interact with energy in pursuit of a more sustainable future.

turnitin Page 47 of 62 - AI Writing Submission Submission ID trn:oid:::1:3441647571

5.3 Project Management and Financial Analysis

An effective project management is key to implement and detect an energy efficiency potential at the beginning as part of IoT-based Smart Energy Meter system. A good planning, a financial study, this making that itself completes in time agreed by with some benefit for those involved parties will be involved. This part describes the elements of project analysis: stages, resources and terms, as well as financial assessment of the cost-revenue-profit structure.

5.3.1 Project Management Overview

Efficient project management is very critical to achieve implementation of IoT based Smart Energy Meter System in its completion time and before its allocated budgeted cost overrun time longevity. Project management style sets ambitious goals, supportive task delegation and constant progress tracking.

- **Project Phases**: The project consists of several phases to maintain an organized approach towards the development, as follows:
    - **Phase 1: Research and Planning**: This phase is 'Early part' of the project, relating to gaining background information about the area to be studied, identifying goals and objectives, platform characteristics (hardware + software) requirements facilitates determining whether or not it is feasible to undertake the development.
    - **Phase 2: Hardware Setup and Integration**: This phase consists of selecting, interfacing and assembling the required hardware components such as the ESP32 microcontroller, PZEM-004T energy meter and verifying their proper operation together.
    - **Phase 3: Software Development**: This is where the system's software is developed including writing of the real-time data capture, transmission and visualization function within the ThingSpeak app.
    - **Phase 4: Testing and Debugging**: Thorough testing is done to validate that all hardware and software are working as expected. All problems are/if found also resolved during check testing.
    - **Phase 5: Deployment and Documentation**: The system is in use, user documentation has been produced and project reports are ready for delivery.

turnitin Page 48 of 62 - AI Writing Submission Submission ID trn:oid:::1:3441647571

- **Resource Allocation**: Hardware resources (ESP32, PZEM-004T), software tools (Arduino IDE, ThingSpeak app) and human resources (developers, project managers, testers) are needed by the project. Rightly distributed makes sure that everyone in the team has what they need and support.

- **Timeline**: The project should have an overall timeline of about 10 weeks with specific milestones and deadlines set out for each phase. Periodic progress reviews will keep the project on course.

5.3.2 Risk Management

It is important to identify and mitigate risks at an early stage so that the project can proceed without any costly surprises. Potential risks and risk mitigation includes:

- **Hardware Compatibility:** ESP32 and PZEM-004T are usually not easy to work together, need debugging. Mitigation can be made by extensive tests during the hardware setup phase and using well-specified components.
- **Wi-Fi Connection Problem:** Link failure might interfere with data transmission. This can be reduced by thoroughly testing the system in different environments and designing it with including options for Wi-Fi range extenders where necessary.
- **Security Issues:** Security is a matter of concern in the system as it involves real-time data transmission. Data breaches will be kept at bay with regular security audits and employing encryption protocols like TLS/SSL.
- **Cost Overruns:** If the project unexpectedly incurs costs, it also may exceed its budget. Keep track of costs frequently, and make changes to resources or time as necessary to keep things under budget.

5.3.3 Financial Analysis

- **Budget Analysis**

The budget of the IoT-based energy meter monitoring system is based on the estimation of hardware devices, cloud services and other overheads. Two budget scenarios are presented:

Component	Quantity	Unit Price (BDT)	Total Cost (BDT)
ESP32 Microcontroller	1	450tk	450tk
PZEM-004T Energy Meter Module	1	1450tk	1450tk
Wi-Fi Connectivity (built-in)	-	-	-
Power Supply & Wiring	1 set	500tk	500tk
ThingSpeak Cloud (Free Tier)	-	-	-
PCB Design & Fabrication (basic)	1	350tk	350tk
Enclosure & Mounting Case	1	500tk	500tk
Miscellaneous (sensors, resistors)	-	400tk	400tk
Total			3650tk

- **Alternate Budget (Cost-Optimized Option)**

Component	Quantity	Unit Price (BDT)	Total Cost (BDT)	Rationale
ESP8266 (NodeMCU)	1	350tk	350tk	Cheaper but not as powerful as ESP32
PZEM-004T Energy Meter Module	1	1450tk	1450tk	Necessity for measuring energy use
Power Supply & Wiring	1 set	350tk	350tk	Basic adapter
Blynk App	-	-	-	Alternative to ThingSpeak
PCB Breadboard (Prototype only)	1	190tk	190tk	Temporary testing, Not for production ensence
Enclosure (3D Printed/Low-cost)	1	350tk	350tk	Reduces case cost
Miscellaneous	-	200tk	200tk	Basic components
Total			2890tk	Cheaper prototype option

- **Rationale for Alternate Budget**

- The alternative design cuts costs by substituting the ESP32 with a cheaper ESP8266, and open source cloud platforms (Blynk, MQTT broker) instead of Thingspeak premium features.
- No PCB has been made, during prototyping this is replaced with breadboard setup.

- While it is a solution that works for small student projects or very limited-sized testing, the low scalability and worse performance are negatives aspects.

➢ Revenue Model

The potential sources of revenue for the IoT energy meter are:

1. **Hardware Sales Model** – selling of the smart energy metering units to homes, industrial or utilities.
    - Expected Sale Price: 8,000-10,000tk per piece
    - Profit Margin per Unit: ~30–40%
2. **Subscription Model** – Providing premium features through cloud services (data storage, notification, energize optimization tips along with third-party IoT assistance).
    - Fallow free mentoring, premium tier at BDT 200-500tk per month.
3. **Collaboration with Utilities** – Offering devices to electric utilities for mass-market deployment and data analysis.

4. **Data-Driven Services** – Data on energy consumption can be operationalized for predictive analytics, demand optimization and consulting.

5.4 Complex Engineering Problem

5.4.1 Complex Problem Solving

The complexity of the design problem and the element integration criterion: From simple component to system, an IoT based Energy Monitoring System development is not a plug-and-play engineering task. It involves a synthesis of understanding multiple branches of engineering in order to compromise the conflicting demands in way that is safe and gives a viable and enduring system. The next map shows why it was considered a hard problem.

Table 5.1: Mapping with Complex Engineering Problem.

EP1 Dept of Knowledge	EP2 Range of Conflicting Requirements	EP3 Depth of Analysis	EP4 Familiarity of Issues	EP5 Extent of Applicable Codes	EP6 Extent Of Stake-holder Involvement	EP7 Interdependence
✓	✓	✓			✓	✓

Mapping with Knowledge Profile for EP1

Table 5.2: Mapping with knowledge Profile.

K3 Engineering Fundamentals	K4 Specialist Knowledge	K5 Engineering Design	K6 Engineering Practice	K8 Research Literature
✓	✓	✓	✓	

5.4.1.1 Justification for EP Attributes Mapping

- **EP1 - Depth of Knowledge:**

This was a project that combined narrow knowledge from several fields:

- ✓ **Electrical Engineer:** The principles of AC power measurement and the interaction with PZEM-004T metrology module.
- ✓ **Electronics & Embedded Systems:** Programmed ESP32 microcontroller, UART communication and Wi-Fi setup.
- ✓ **Computer Science & Networking:** Configuring IoT device-to-cloud connectivity and using the Thingspeak API to process data.
- ✓ **Software Integration:** To design a system architecture to integrate hardware sensor/firmware logic/mobile application visualization.

The challenge is on how to interconnect these heterogeneous knowledge domains for developing a reliable and self-explicative monitoring system.

- **EP2 - Range of Conflicting Requirements:**

This project required the balance of several important engineering trade-offs:

- ✓ **Accuracy vs. Cost:** The use of the pre-fab module directly addressed the reliability of accuracy, where it locked horns with the foundations-approach to in-depth learning that a custom metrology design would offer.
- ✓ **Differences:** realtime data vs. power consumption: The ESP32 Wi-Fi gave live data but was not compatible for low-power, long term operation.
- ✓ **Ease of Use (Thingspeak) vs. Customization:** Although the Thingspeak platform allowed for quick development, it contradicted system flexibility, data ownership and control over advanced features.

- **EP3 - Depth of Analysis:**

This task involved multilevel analysis:

- ✓ **Hardware Level:** Monitoring the UART communication protocol between ESP32 and PZEM-004T for correct data parsing and error recovery.
- ✓ **System-Level:** Taking the complete pipeline from sensor measurement through cloud visualization and identifying spots like network latency, or data packet loss.
- ✓ **Software Level:** Studying the API limitations and structure of Thingspeak to have accurate mapping of sensor data to channel fields, thus also making the app dashboard more stable

- **EP6 - Extent of Stakeholder Involvement and Conflicting Needs:**

The challenges of competing stakeholders with potentially conflicting priorities have been coped with in this project:

- ✓ **End-Users/Consumers:** Their main requirement is user friendly and easy to navigate mobile interface (specifically a Thingspeak app) that gives them clear, factors to act on the energy consumption data of their house so they can save dollars.
- ✓ **Utility Providers / System Operators:** For reliable, accurate data collection for billing or grid monitoring, robust hardware with secure and uninterrupted data transmission is the primary concern which may add in system complexity and cost.
- ✓ **Developers / Engineers:** The goal of developing a maintainable, scalable and technically elegant system may be antithetical to the consumer request for a simple solution and budget limitations on cheap hardware such as the ESP32 and PZEM module.

- **EP7 - Inter-dependence:**

Interdependency among the components of the system:

- ✓ **Hardwired Firmware:** The firmware running on the ESP32 was command driven by the UART protocol of PZEM module.
- ✓ **Cloud Dependent:** Features like ThingSpeak app was entirely depended on ESP32 to be connected to Wi-Fi with a stable connection and format data correctly for its API.
- ✓ **Design Cascade:** Design choices were made early in the process of development, for example selection of PZEM module eliminated need to design deep analog circuit but ESP32 Wi-Fi as communication protocol compromised due to network stability and size of power consumptions. Should any of the parts change (re: using Lora for communication), This would throughout a prototype-built-for-production change, away alongside with modification and/ or rewriting of firmware and cloud.)

turnitin Page 53 of 62 - AI Writing Submission Submission ID trn:oid:::1:3441647571

5.4.1.2 Justification for Knowledge Profile Mapping (linked to EP1):

- **K3 - Engineering Fundamentals:**

This effort leveraged basic engineering principles:

- ✓ **Electrical Principles:** Applied AC circuit theory, power (P=IV) and energy (kWh) calculation to deduce sensor data.
- ✓ **Electronics:** Supported ESP32 and PZEM-004T integration through UART communication.
- ✓ **Software:** Designed firmware for task scheduling, data processing and Wi-Fi network incorporation.

Networking: We put the fundamental IoT principles into action by connecting a device to the Cloud through Wi-fi and using APIs for data transfer.

- **K4 - Specialist Knowledge:**

The specific expertise on which this project was based included:

- ✓ **IoT Protocols:** Hands on experience with Wi-Fi and MQTT/HTTP for device to cloud communication.
- ✓ **Sensor Integration:** Experience interfacing with the proprietary UART protocol of the PZEM-004T power monitoring module.
- ✓ **Cloud Platforms:** Domain knowledge of Thingspeak API, fields in channels and design of mobile dashboard.
- ✓ **Embedded Renderer for ESP32:** Lower level programing of the ESP32 platform, managing power and peripherals.

- **K5 - Engineering Design:**

Engineering design was utilized through this work with:

- ✓ **System Architecture:** Design of the whole data flow from sensor measurement (PZEM) to user interface (Thingspeak), combining hardware and software.
- ✓ **Component Selection:** Informed selection of the ESP32 for processing/Wi-Fi and PZEM module to ensure metrology accuracy.
- ✓ **Interface Design:** Structuring the ThingSpeak dashboard to show energy data in a clear and intuitive way for the user.
- ✓ **Firmware Development:** Developing organized code that will monitor sensor communication, data processing and network connection in a reliable way.

- **K6 - Engineering Practice:**

This initiative applied fundamental engineering through:

turnitin Page 54 of 62 - AI Writing Submission Submission ID trn:oid:::1:3441647571

- ✓ **Interactivity:** From an abstract concept to a reality, getting the fingers of users dirty through implementing workforce feedback on its way.
- ✓ **Component Sourcing & Integration:** Selecting, purchasing and physically integrating the hardware parts (ESP32, PZEM- 004T, wires).
- ✓ **Testing & Validation:** Validating the correctness of meter readings against a trusted source and testing system reliability under various network conditions.
- ✓ **Documentation:** Drafting clear documentation about the design, code, and setup flow in order to provide ease for other individuals to go for replication or future maintenance.

5.4.2 Engineering Activities

Table 5.3: Mapping with Complex Engineering Activities.

EA1 Range of re-sources	EA2 Level of Interaction	EA3 Innovation	EA4 Consequences for society and environment	EA5 Familiarity
✓	✓	✓	✓	

5.4.2.1 Justification for Engineering Activities Mapping:

- **EA1 - Range of Resources:**

This assignment involved choosing and working with a variety of sources:

- ✓ **Hardware Resources:** ESP32 microcontroller, PZEM-004T sensor module and wires, power sources.
- ✓ **Software resources:** Arduino – IDE (used to program the ESP32), ThingSpeak platform (for app designing and cloud service), library for PZEM electric sensor, and Wi-Fi.
- ✓ **Information Resources:** Datasheets for the ESP32, PZEM-004T, API Documentation for Thingspeak and online technical forums.
- ✓ **Test Resources:** Verification by multimeter and system verification with real-world electrical loads.

- **EA2 - Level of Interaction:**

This was a complex project which involved many technical and non-technical dimensions:

- ✓ **Technical Integration:** Strong bond of the hardware (ESP32, PZEM), software (firmware, Thingspeak API) and network (Wi-Fi).
- ✓ **Stakeholder Engagement:** How to design to service both users (an easy, enjoyable app interface) and administrators (satisfy actual data).
- ✓ **Platform Interaction:** Reliability on 3 rd Party platforms (Thingspeak) to carry out operations which required the tailoring of our code to their respective specifications and APIs.
- ✓ **Debug-Loop Interaction:** Recurrent bug hunting across the hardware wiring, firmware software and the network connections to get the system up-and-running with acceptable behavior.

- **EA3 - Innovation:**

This project demonstrated innovation through:

- ✓ **System integration:** New use of OTS (ESP32 + PZEM) with the Thingspeak platform to make a working low-cost energy monitoring solution.
- ✓ **Accessibility Design:** Developing a user-friendly mobile interface for non-technical personnel to use advanced energy content.
- ✓ **Simple IoT application:** Develop a workable IoT in action- e.g., an energy saving at house system may works for kind of smart houses.
- ✓ **Customization:** Making and program the generalized HW (ESP32) for a metering and monitoring application.

- **EA4 - Consequences for Society and Environment:**

This initiative is actively pursuing positive societal and environmental benefit by:

- ✓ **Energy awareness:** Give users fast, real-time usage information so they can make informed decisions and reduce consumption –and carbon footprint!
- ✓ **Save Money:** Assists families and businesses identify wasteful usage so you save on utility bills.
- ✓ **Accessibility:** Offer a cost-effective, scalable monitoring system that covers multiple types settings.
- ✓ **Smart Infrastructure:** Which plays a role in building the smarter and efficient energy grids, sustainable resource management.

turnitin Page 56 of 62 - AI Writing Submission Submission ID trn:oid:::1:3441647571

5.5 Summary

The proposed Smart Energy Meter powered by IoT is aimed at effective, environmentally friendly and user-friendly energy consumption monitoring and controlling. The plant provides opportunity for users on individual or small group basis to reduce the consumption of energy and cut down on electricity bills, and help the environment by being an advocate for energy efficiency. The project is broken down into distinct steps including research/planning, setting up hardware, developing software, testing and deployment. Successful project management techniques and good practice help to develop the system on time, within budget and with low risk. The financials illustrate the system's profit potential, which includes monetizing through product sales, subscriptions and partnerships. Initially it is hardware/software/labor overhead, but this system can grow and has long term potential. Strategies such as customer retention, product updates and support services will enable sustained revenues and users satisfaction. The project also establishes a plan that is consistent with energy sustainability objectives by decreasing the amount of wasted energy, increasing integration of renewables and adopting sustainable design practices and materials. Ethics The system complies with data privacy and security regulations, to safeguard and manage user information responsibly. It is also inclusive in that it makes the system inexpensive and available to as wide a range of users as possible. The sustainability plan guarantees the system's long-term viability both technically and environmentally, contributes to global energy conservation and sustainable development.

In summary, the IoT-based Smart Energy Meter system provides numerous advantages for users, enterprises and planet earth; hence is a hopeful state-of-the-art technology due to energy management. They are simple words, but, functioning as they should in rhetoric, they have combined to give a political innovation that comes unashamedly packaged and is one of the benign pesticides used in our abiotic spectacle - The Smart City.

Chapter 6

Conclusion

This chapter describe the project's Summary and Limitations. It also discusses the Future work of this project.

6.1 Summary

This chapter finally summarizes the Summary and Limitations of the project before exploring its Future work. 6.1 Summary The IoT-based Smart Energy Meter system is an innovative solution that addresses the growing demand for energy efficiency and sustainability. The system allows users to monitor their energy consumption in real-time, enabling them to take more informed decisions about avoiding waste and optimizing energy usage. This results in not only financial savings for households and corporations but also environmental sustainability, as it lowers carbon emissions and promotes energy conservation. With good project management and financial planning, the system will be developed and implemented within the specified timeframe and budget. The ease of scalability and opportunity for generating revenue, as well as the good it does to society and the climate, makes the device a valuable investment. Focused attention to user access, data information security, and moral considerations makes this device useful to a broader audience, from low-income households to organizations seeking to boost energy efficiency. In addition, the device's sustainability plan assures long-term viability and a dedication to constant excellence and resource conservation as the global energy sector moves to renewables. The IoT-based Smart Energy Meter system is thus a promising solution for the future in combining innovation with a solid commitment to sustainability that adds extra boosts to the system, not just to the user.

6.2 Limitation

The IoT based Smart Energy Meter system is beneficial in many ways; however, it does have its own set of limitations. Its reliance on a strong Wi-Fi connection for live data looks to be its biggest drawback. Users in areas of sparse Wi-Fi coverage may see a time lapse or halt in data updates, reducing the efficacy of the system. Moreover, the accuracy of either energy data may be dependent on quality of one or more electrical components and calibration of the system, which is typically not in compliance with satisfactory standards

for official energy measurement. Power draw of the system (though kept minimum using energy-efficient ESP32 microcontroller) can be an issue for off-grid or low power consumers. In addition, the scalability can be problematic when you integrate several energy meters in bigger systems, since if the system does not scale simply definite quality of service will decrease (i.e. data congestion and performance) as you increase bittering devices. Although the unit is meant to be relatively user-friendly, it will take some level of technical know-how to install and troubleshoot which might alienate technically challenged users. Compatibility Drawbacks –Further, compatibility with legacy systems linked to its target application environment; namely, modern IOT is another limitation of the system that are discussing- and might not be seamless compared to what it can offer within modern environments. Finally, anywhere official energy measurement or billing would be taking place, regulatory compliance could become a problem in some jurisdictions. Mitigating these limitations with additional development and optimization will be necessary to facilitate widespread adoption of the system, and support its long-term sustainability. However, despite the abovementioned challenges, the technique is a promising approach by which to enhance energy efficiency and aid environment sustainability.

6.3 Future Work

Smart Energy Meter over IoT system has robust scope for future extension and performance improvement further scalability. Scalability is one important area that needs improvement, especially in case of larger scale setups where more than one energy meters are used. Future work will develop ways to scale the system infrastructure in order to handle data from multiple devices without hindering performance. Furthermore, to enhance the accuracy and calibration of energy readings is important so that the system can satisfy the official energy measurement standards in order for the system to have additional applications in billing and regulation. Another significant domain for improvement is the integration of renewable energy systems. With the growing share of green energy sources such as solar and wind in mind, it might also be improved to measure and control the production of those energy sources, enabling users to schedule their power usage against a green energy supply. Minimization of power requirement of the system, especially for off-grid users, will also be a focus. By implementing some energy-saving modes and using solar-based solutions, the system can be more suitable for low power or no-power grids. In the future, we will be considering to integrate advanced analytics and

AI into the system that could provide personalized energy conservation suggestions and predictive maintenance alerts according to data of users. Adding additional communication interfaces, such as Lora WAN or 5G, could increase the flexibility of this system to broader circumstances and overcome its capacity limitations. Improving the UI of mobile App, Adding voice control, and more languages are also available for multi-user and multi-people diggings.

Also critical is that the system complies with regulation and earns certifications of accuracy and safety needed to widen such a systems use in official energy measurement and billing. 4) At last, the system's market penetration should be grown, specifically in remote areas with low smart energy technology accessibility. Through system tailoring to suit local conditions, and joint development with utilities and governments, the solution is able to be scaled around the world in support of global sustainable energy management.

turnitin Page 60 of 62 - AI Writing Submission Submission ID trn:oid:::1:3441647571

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